

Ocean's 11 Golf App

Technical Design Document (Aligned to PRD v1.0)

Status

Approved for Development

This document implements the requirements defined in the Product Requirements Document (PRD) Version 1.0.

1. Architecture Overview

Ocean's 11 is a local-first mobile application.

The application is designed for:

- Single-device use
- No user accounts
- No cloud dependency
- No round history

Primary objective:

Fast and accurate Ocean's 11 scoring during a live round of golf.

2. System Architecture

Client:

React Native Mobile Application

Database:

SQLite

Storage Model:

Local Device Storage

Architecture:

Local-First

No backend required for V1.

3. Data Persistence Strategy

Persistent Data

Persist between application launches.

Course

CourseHole

Courses remain available indefinitely.

Temporary Data

Exist only during an active round.

Round

RoundPlayer

HoleScore

Upon completion:

Round data is discarded.

No historical round storage.

4. Data Model

Course

Fields:

CourseID (UUID)

CourseName

TeeName (Nullable)

CreatedDate

CourseHole

Fields:

CourseHoleID (UUID)

CourseID

HoleNumber

Par

StrokeIndex

Validation:

- Exactly 18 holes
 - Stroke indexes 1–18
 - No duplicate indexes
-

Round

Fields:

RoundID

CourseID

CurrentHole

Status

Status Values:

- Active

- Complete
-

RoundPlayer

Fields:

RoundPlayerID

PlayerName

CourseHandicap

HoleScore

Fields:

RoundPlayerID

HoleNumber

GrossScore

Counted

AutoCounted

No calculated values stored.

5. Calculation Engine

All scoring calculations are derived.

No totals are permanently stored.

Handicap Stroke Allocation

Inputs:

CourseHandicap

StrokeIndex

Formula:

$\text{base_strokes} = \text{floor}(\text{course_handicap} / 18)$

$\text{extra_strokes} = \text{course_handicap} \% 18$

$\text{strokes_received} = \text{base_strokes}$

if $\text{stroke_index} \leq \text{extra_strokes}$

$\text{strokes_received} += 1$

Net Score

$\text{net_score} = \text{gross_score} - \text{strokes_received}$

Result Versus Par

$\text{result} = \text{net_score} - \text{par}$

Display Values:

-2

-1

E

+1

+2

Ocean's 11 Score

Ocean11Score =

$\text{sum}(\text{result versus par on counted holes})$

Lower score wins.

6. Ocean's 11 Rules Engine

Maximum Count Rule

Counted holes may never exceed 11.

Attempting to count a 12th hole generates validation error.

Automatic Count Rule

Definitions:

Counted = Current counted holes

Needed = $11 - \text{Counted}$

Remaining = Unplayed holes remaining

If:

Needed = Remaining

Then:

All remaining holes become AutoCounted.

Auto Count Editing Rule

Auto-counted holes may be modified only if:

Player can still mathematically reach exactly 11 counted holes.

7. Recalculation Engine

The application recalculates whenever:

- Gross Score changes
- Counted status changes
- Course Handicap changes

Execution Order:

1. Remove AutoCounted flags
2. Recalculate handicap strokes
3. Recalculate net scores
4. Recalculate results
5. Recalculate counted totals
6. Reapply auto-count rules
7. Recalculate Ocean's 11 scores
8. Recalculate leaderboard

All displayed values are derived.

No stale values permitted.

8. Screen Architecture

Home Screen

Components:

- New Round
- Courses

No additional navigation.

Courses Screen

Components:

- Search
- Course List
- Create Course

Actions:

- Create
 - Edit
 - Delete
-

Create/Edit Course Screen

Fields:

Course Name

Tee Name

Hole Grid

Columns:

Hole

Par

Stroke Index

Actions:

Save

Cancel

Round Setup Screen

Components:

Course Selector

Player List

Fields:

Player Name

Course Handicap

Actions:

Add Player

Remove Player

Start Round

Round Screen

Primary application screen.

Designed so users can complete an entire round without leaving this screen except to correct mistakes.

Header

Course Name

Current Hole

Par

Stroke Index

Official Leaderboard

Always visible.

Columns:

Player

Score

Counted

Status

Status Values

Need X

Auto

Locked

Example

Fran | -2 | 6/11 | Need 5

Bob | +1 | 7/11 | Need 4

Steve | E | 8/11 | Auto

Mike | -1 | 11/11 | Locked

Score Entry Grid

Columns:

Player

Gross

Result

Count

Result is read-only.

Count is toggleable.

Save Button

Save Hole

Actions:

1. Validate
2. Recalculate
3. Apply auto-count logic
4. Update leaderboard
5. Advance to next hole automatically

No Next Hole button exists.

Scorecard Access

Scorecard icon only.

Used exclusively for editing previous holes.

Scorecard Screen

Purpose:

Review and edit previous holes.

Not used for leaderboard review.

Not used for score tracking.

Layout

Hole

Gross

Result

Count

Editing

Tap hole row.

Open bottom-sheet modal.

No separate Edit Hole screen exists.

Edit Modal

Displays:

Hole Number

Par

Stroke Index

Gross Score

Result

Count Status

Editable:

Gross Score

Count Status

Actions:

Save

Cancel

Save Behavior

Immediately triggers full recalculation engine.

Return to Scorecard.

Return To Current Hole

Button available.

Returns user directly to active hole.

Final Results Screen

Displays:

Final Ocean's 11 Leaderboard

Example:

1. Fran -4

2. Bob -1

3. Steve +2

4. Mike +3

Action:

Finish Round

Round data discarded.

Return Home.

9. Acceptance Criteria

The build is accepted when:

- ✓ Courses can be created
- ✓ Courses can be edited
- ✓ Courses can be deleted
- ✓ Rounds can be created
- ✓ Gross scores can be entered
- ✓ Counted holes can be selected
- ✓ Handicap allocation is correct
- ✓ Ocean's 11 calculations are correct
- ✓ Auto-count logic is correct
- ✓ Leaderboard updates instantly
- ✓ Hole edits recalculate all downstream values
- ✓ Save automatically advances to next hole
- ✓ Users can complete a round without opening Scorecard except for corrections
- ✓ Final results display correctly

✓ No manual calculations are required